Efficient Data Structures for
RDD-based Apriori on Spark

Luis Alberto Vásquez Vargas
IC – UNICAMP

MO809 – Advanced Topics in Distributed Systems
Instructor: Prof. Luiz Fernando Bittencourt

l322989@dac.unicamp.br
April 7, 2026

Outline

1. The Base Study
2. Introduction & Motivation
3. The FIM Problem
4. Infrastructure Evolution
5. Literature Overview
6. Proposed Data Structures
7. Experimental Results
8. Conclusion
9. References

The Base Study

Original Research Paper

A Data Structure Perspective to the RDD-based Apriori Algorithm on Spark

- **Authors:** Pankaj Singh, Sudhakar Singh, P. K. Mishra, and Rakhi Garg.
- **Journal:** *International Journal of Information Technology* (Springer), 2019.
- **Focus:** Efficiency of **Hash Tree**, **Trie**, and **Hash Table Trie** in Spark.

Introduction & Motivation

- **The Goal:** Discovering hidden patterns in massive transactional datasets.
- **The Barrier: Combinatorial Explosion**
  - For a set of m items, the search space is $O(2^m)$.
  - If $m = 100$: $2^{100} \approx 1.26 \times 10^{30}$ combinations.
  - **Brute Force:** Checking every subset would take longer than the age of the universe.

Introduction & Motivation

- **Hardware Evolution:**
  - We have the "Ferrari" of computing: **Apache Spark RDDs**.
- **Software Bottleneck:**
  - Legacy structures (Hash Trees) create a **traffic jam** that wastes the speed of RAM.

The Core Thesis

Infrastructure alone cannot solve $O(2^n)$. To exploit the speed of Spark, we must replace "Disk-era" structures with memory-optimized **Tries**.

The FIM Problem

- **Frequent Itemset Mining (FIM):** Finding subsets with frequency $\geq$ *minsup*.
- **The "Bundle Deal" Analogy:**
  - If customers buy GPUs, what else do they buy?
  - $n = 1000$ items lead to 2^{1000} possible combinations.

- **The Apriori Principle:**
  "Any subset of a frequent itemset must also be frequent."
- We prune the space, but we need efficient storage for millions of candidates.

Infrastructure Evolution

Coming Soon...

Coming Soon...

Proposed Data Structures

Coming Soon...

Experimental Results

Coming Soon...

Conclusion

Coming Soon...

References

P. Singh, S. Singh, P. K. Mishra, and R. Garg.
A data structure perspective to the RDD-based Apriori algorithm on Spark.
Int. J. Inf. Technol., 14(3):1585–1594, 2019.

Questions?

Luis Alberto Vásquez Vargas
IC – UNICAMP

l322989@dac.unicamp.br